

100 Example BI Queries for a Retail OLAP Cube

Below is a list of 100 example Business Intelligence (BI) questions that an enterprise data analyst at a large retailer (like Home Depot) might ask an advanced AI-powered analytics system. These cover key retail domains – sales, inventory, store operations, e-commerce, and product merchandising – reflecting common metrics and analyses in retail BI ¹ ² . The queries range from straightforward data retrieval (e.g. total sales figures) to deeper analytical tasks (such as forecasting and clustering) and even requests for visualizations. This diversity showcases how a sophisticated AI agent (equipped with tools like pandas, scikit-learn, statsmodels, matplotlib, etc.) could be directed to extract insights from a retail OLAP cube.

Sales & Revenue Analysis

1. What were the total sales in each region last quarter, and how did this compare to the same quarter last year?
2. Which product category had the highest year-over-year sales growth, and by what percentage did it grow?
3. What are the top 10 products by revenue this month, and how do their sales compare to last month's top 10 products?
4. Show the trend of monthly sales and gross profit over the past two years, broken down by major product category (visualizing any seasonal peaks or dips).
5. Which stores or regions contributed most to the increase in sales this year, and which ones lagged behind in growth?
6. What is the average transaction value (average sale per customer) and how has this metric changed over the last year? ³
7. How do weekend sales compare to weekday sales on average, and are there specific regions or stores where this difference is especially pronounced?
8. How do actual sales year-to-date compare to the sales targets or forecasts for the same period? Identify any significant gaps.
9. Identify the revenue contribution of each product category as a percentage of total sales, and determine which categories are driving overall performance.
10. Which stores or sales regions exceeded their sales targets last quarter, and which fell short of their targets (and by what margin)?
11. What is the year-to-date overall sales growth rate, and how does it break down by product category or region (e.g., which categories are growing or declining)?
12. How did a specific promotion (e.g., a 20% off summer sale) affect the sales of the promoted products or category during the promotion period versus before and after?
13. Calculate the gross profit margin for each product category and identify any categories where the margin is shrinking over time.
14. What are the monthly sales figures and gross profit for our e-commerce channel versus our physical stores, and how have these trends changed over the last 12 months?
15. Which customer segment (e.g., DIY customers vs. professional contractors) generates the highest revenue, and how do their purchasing patterns differ?

16. What is our customer retention rate or repeat purchase rate this year, and how does it compare to last year? (For example, what percentage of last year's customers made another purchase this year?)
17. Use a time-series forecasting model to predict next quarter's sales for each major category based on the trends from the past three years.
18. Compare the holiday season sales (e.g., November-December) for each of the last five years to see how our peak season performance is trending year-over-year.
19. Plot a heatmap or similar visualization of sales by month (rows) and region (columns) to highlight seasonal patterns and regional differences in performance.
20. Determine the sales per square foot for each store, and identify which stores are the most and least productive in terms of sales per floor area ⁴ .

Inventory & Supply Chain Analysis

1. Which products are at risk of stocking out soon based on current inventory levels and recent sales velocity (i.e., fast-selling items with low stock)?
2. What is the inventory turnover rate for each product category over the past 12 months, and which categories have the fastest vs. slowest turnover? ⁵
3. Identify the top 10 products with the highest inventory turnover and the 10 products with the lowest turnover (indicating slow-moving stock).
4. Which items have been in inventory for over 90 days (slow-moving or aging inventory), and what is the total value of this excess stock?
5. What is the total inventory value on hand for each product category, and how does the current value compare to last quarter's inventory value?
6. How many days of supply do we currently have for each major product category (based on current stock levels and average daily sales)?
7. Which products experienced stockouts in the last month, and approximately how many sales were lost due to those stockouts (estimating unmet demand)?
8. Analyze the trend of inventory levels versus sales for Category X over the last year – are inventory levels keeping up with sales, or do we see periods of overstock/understock?
9. What is the rate of product returns for each category, and which products have the highest return rates? How might these returns be impacting net sales? ⁶
10. Perform an ABC analysis on our product inventory to classify items into A, B, and C categories based on their sales revenue contribution ⁷ .
11. Which SKUs (stock-keeping units) account for roughly 80% of our total sales? (Identify the "A" items that drive the bulk of revenue.)
12. Are there products we should consider discontinuing due to consistently low sales and high inventory levels (i.e., products that tie up capital but don't sell)?
13. How do current inventory levels compare to the optimal or target stock levels for each product? Highlight any significant overstocks or understocks.
14. What is our order fill rate (the percentage of customer orders that were fulfilled completely) in the last quarter, and which products or categories had the most backorders or unfulfilled orders?
15. How many backorders did we have in the past month, and which products are most commonly backordered or delayed due to stock shortages?
16. Compare this year's inventory turnover by category to last year's – which categories have improved and which have slowed down in how quickly they sell through inventory?
17. Which suppliers have the longest lead times or most frequent late deliveries, and how are these supply issues affecting our inventory levels or stockouts?

18. Analyze inventory shrinkage (loss due to theft, damage, or errors): what is the shrinkage rate overall and are there particular stores or categories with higher shrinkage?
19. Visualize the current inventory levels across all warehouses or stores (for example, a bar chart by location) to see where most of our stock is concentrated.
20. Using sales forecasts, predict the inventory requirements for the next quarter for our top-selling products to ensure we meet demand without overstocking.

Store Operations & Performance

1. Which store had the highest foot traffic last quarter, and did that location's high traffic translate into equally high sales? (Analyzing if more visitors lead to more sales.)
2. What is the conversion rate at each store (the percentage of store visitors who make a purchase), and which stores have the highest and lowest conversion rates? ⁸
3. Identify any stores that saw a significant decline in sales compared to the same period last year. What potential factors could explain the drop (e.g. increased competition, local economic changes)?
4. How do sales patterns differ between weekdays and weekends for each store? For example, are there stores that do more business on weekends versus weekdays, or vice versa?
5. Cluster the stores based on performance metrics (like total sales, sales growth, and inventory turnover) to find groups of similar stores – for instance, to group stores into high performers, mid-range, and low performers.
6. Which stores have the highest sales per employee, and what practices might be contributing to their higher productivity? ⁹
7. Are there specific regions where all the stores are underperforming in sales (relative to other regions), suggesting broader regional challenges?
8. How did the stores that underwent a recent remodeling or floor layout change perform afterward, compared to stores that did not? Did remodeling lead to a sales lift?
9. How do major weather events affect our store sales? For example, did stores in regions that experienced a snowstorm or hurricane see a dip (or spike) in sales during that period?
10. Which stores have the highest rate of product returns or exchanges, and what products or categories are most frequently returned at those locations?
11. At what times of day does each store experience peak sales, and are staffing levels aligned with those peak hours to avoid missed sales opportunities during busy periods?
12. Is there a correlation between customer satisfaction (e.g., survey scores or Net Promoter Score) and sales at the store level? Do stores with higher satisfaction scores also see higher sales?
13. Identify stores that frequently run out of certain high-demand products. How often do these stockouts occur, and how might they be impacting sales at those stores?
14. Analyze the impact of opening a new store on the sales of nearby existing stores. For example, after opening Store X, did the sales at neighboring Store Y change significantly (potential cannibalization)?
15. Does store size (in square feet) correlate with its sales performance or profitability? For instance, do our larger stores consistently generate more sales, or are some small-format stores very efficient?
16. What portion of each store's sales comes from fulfilling online orders (e.g., Buy Online Pickup In-Store or ship-from-store), and how is this influencing store labor and inventory needs?
17. Detect any anomalies in weekly sales by store – for example, stores that had an unexpected sales spike or drop in a given week – and investigate possible reasons (like local events or data errors).
18. Which stores achieved the highest year-over-year sales growth, and what are those stores doing differently (in terms of marketing, staffing, etc.) that might be contributing to their success?

19. Compare the effectiveness of recent sales promotions across stores. Did some stores capitalize on a promotion (like a weekend sale) more effectively than others, and what factors might explain the difference?
20. Evaluate how a special in-store event (e.g., a promotional workshop or a grand reopening event) impacted that store's performance. Look at the store's sales and foot traffic during the event and in the weeks after to gauge lasting effects.

Online & E-commerce Performance

1. What is the overall conversion rate of our e-commerce website (the percentage of website visits that result in a purchase), and how has this conversion rate changed month over month?
2. How many unique visitors did our online store receive last month, and what percentage of those visitors ended up making a purchase (i.e., the conversion rate for that month)?
3. Which online traffic sources (such as organic search, paid ads, email campaigns, or social media) are bringing in the most visitors, and which sources yield the highest conversion rates versus others?
4. What is the average order value for online sales, and how does it compare to the average order value for in-store sales? (This could highlight differences in customer purchase behavior online vs. offline.)
5. Analyze the e-commerce conversion funnel: out of all website visitors, how many proceed to view a product, how many add a product to cart, and how many ultimately complete a purchase? Identify where the biggest drop-offs occur in the funnel.
6. Which product categories perform best in the online channel compared to in-store? For example, are there categories that have much higher online sales as a proportion of total sales?
7. What is the shopping cart abandonment rate on the website (percent of carts created that don't result in purchase), and did any recent website changes or promotions affect this metric? ¹⁰
8. How do conversion rates differ between mobile users and desktop users on our website? For instance, is the mobile conversion rate significantly lower, and if so, what might be the cause?
9. During what time of day does our website see the highest sales volume, and are there particular times that have a lower conversion rate (which might indicate performance issues or other problems)?
10. Which online marketing promotions or discount codes were most effective in driving sales, and what was the sales lift or ROI for those campaigns?
11. What percentage of online orders are returned by customers, and how does this online return rate compare to the return rate for in-store purchases?
12. How has the introduction of free shipping (or changes to the free shipping threshold) impacted our online conversion rate and average order value? (Compare before vs. after the change.)
13. What is the geographic distribution of our online sales? Identify the regions or cities with the highest online sales volumes, and consider if our marketing and logistics align with those areas.
14. What percentage of online orders are picked up in a physical store (BOPIS), and how has this ratio changed over the last year? Are customers increasingly using the buy-online-pickup-in-store option?
15. Build a predictive model (e.g., a logistic regression) to predict which website visitors are most likely to make a purchase, based on their browsing behavior (such as pages viewed, time on site, etc.).
16. Use clustering to segment our online customers based on their behavior and spending (for example, into groups like high-value repeat customers, occasional bargain shoppers, one-time purchasers, etc.), and describe the characteristics of each segment.
17. What is the average customer lifetime value (CLV) for customers acquired through online channels, and how does it compare to the CLV of customers who primarily shop in-store?

18. What are the most common search terms that users enter on our website, and how often do searches for those terms convert into actual sales? (This could reveal if customers find what they're searching for.)
19. Compare the online sales on major shopping days (such as Black Friday or Cyber Monday) to the sales on regular days. How much of a spike do we see, and which product categories saw the biggest jumps during those events?
20. Do products with higher customer ratings (or better reviews) on our website sell significantly more than products with lower ratings? Analyze the relationship between average rating and sales volume for products.

Product & Merchandising Analysis

1. What percentage of total revenue does each major product category contribute? (Identify our biggest revenue drivers in terms of product categories.)
2. Which subcategories (or product groups) within our major categories are growing the fastest in sales, and which are shrinking? (This highlights emerging trends within the product hierarchy.)
3. Which brands are our top performers in terms of sales, and which brands are underperforming? Provide a ranking of brand performance within a key category (e.g., top brands in Tools category by sales).
4. For new products introduced in the last year, how did their sales trend over time? (For example, did they have a strong launch and then taper off, or grow steadily? This examines product launch performance over 3, 6, 12 months.)
5. Which products have the highest profit margins, and how do those high-margin products' sales compare to products with lower margins? (Are our high-margin items also our best sellers or not?)
6. Identify any products that have experienced a decline in sales for at least three months in a row. These could be products falling out of favor that might need attention (marketing, discounts, or discontinuation).
7. Which products have the highest return rates (as a percentage of units sold), and what are the common reasons for returns if we have that information? (This could signal quality or suitability issues with the product.)
8. Perform a market basket analysis to discover which products are most frequently purchased together. Based on these findings, suggest product bundles or cross-selling opportunities.
9. What is the attachment rate of accessories or add-on products for our big-ticket items? For instance, when someone buys a power drill, how often do they also purchase related accessories like drill bits or batteries in the same transaction?
10. How have recent price changes affected sales volume for certain products? For example, if we lowered the price of Product X by 10%, did its sales units increase significantly, or did revenue per week change noticeably? (Assess price elasticity.)
11. Which five products had the largest increase in sales (in dollars or units) compared to last year, and which five products saw the biggest sales decline? What might explain these changes?
12. How do our private-label (store brand) products perform versus national brand products in the same category? Compare their sales volumes and profit margins to see if our own brands are capturing significant share.
13. What seasonal sales patterns exist for our products? For example, which categories or products peak in sales during summer or holiday season, and which sell consistently year-round? (Identifying seasonal vs. evergreen products.)

14. Did the introduction of a new product (Product X) cannibalize the sales of an existing product (Product Y) in the same category, or did it expand overall category sales? ¹¹
15. Cluster our products based on their sales performance and profitability metrics (e.g., sales volume, growth rate, profit margin) to identify groups like “high-volume low-margin” products vs. “low-volume high-margin” products, etc.
16. Use a regression analysis to determine which factors most strongly drive a product’s sales. Consider variables such as the product’s price, its average customer rating, how often it’s promoted, and its in-stock rate to see what correlates with higher sales.
17. Are there products that get a lot of online page views or clicks but have a low conversion to actual sales? Identify these “browse but not buy” products – they may need improved descriptions, reviews, or a price adjustment.
18. After the peak season ends, how well do seasonal products continue to sell, and do clearance discounts effectively clear out the remaining stock? (For example, how did summer seasonal items sell in the off-season with markdowns applied?)
19. How has the assortment of products in each category changed over time (e.g., number of SKUs), and did changes in the breadth of assortment impact sales or customer satisfaction? (This can reveal if expanding or pruning the product range had effects.)
20. Calculate the Gross Margin Return on Investment (GMROI) for each product category to evaluate which categories are generating the most profit relative to the inventory investment in them ⁴.

Sources: The above scenarios draw upon typical retail analytics metrics and practices. Key retail performance indicators include metrics like conversion rate, foot traffic, sales per square foot, and inventory turnover ². Retail BI analyses span inventory (e.g., turnover, returns) ⁶, sales trends, e-commerce funnel metrics (like cart abandonment) ¹⁰, and product portfolio management (e.g., ABC classification ⁷ or cannibalization analysis ¹¹). These examples are informed by industry-standard KPI guides and retail analytics resources.

-
- ¹ ⁶ ¹⁰ 25 Retail KPIs & Metrics to Track | NetSuite
<https://www.netsuite.com/portal/resource/articles/financial-management/retail-kpis.shtml>
- ² ³ ⁴ ⁵ 8 Important Metrics for Retail Industry KPIs | Tableau
<https://www.tableau.com/learn/articles/retail-industry-metrics-kpis>
- ⁷ ABC Analysis: How to Analyze Product & Inventory (2024) - Shopify
<https://www.shopify.com/retail/abc-analysis>
- ⁸ ⁹ 12 Critical Retail Industry Performance Metrics (2025) | Retalon
<https://retalon.com/blog/retail-industry-performance-metrics-kpis>
- ¹¹ Product Cannibalization in Retail (Definition, Examples, Tips) | Retalon
<https://retalon.com/blog/product-cannibalization>